LIMKOKWING UNIVERSITY OF CREATIVE TECHNOLOGY

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DIWA2110 • Web Application Development

Group Assignment 40%

Faculty : Information & Communication Technology
Progam/class Name : Diploma in Information Technology
Diploma in Information Technology
Semester : 3
Commence Date : Week 5
Deadline Date : Week 14
Marks : 93

Assignment: Full-Stack Development

Batloung Fruits purchases fruits from suppliers located in different provinces of South Africa. Since fruit availability varies according to season and supplier, the company may purchase apples from one province, bananas from another province, and oranges, mangoes or grapes from other provinces.

For each supplier, Batloung Fruits needs to maintain information such as supplier name, province, contact person, telephone number, email address, and physical address. The system should recognise all nine provinces of South Africa; Eastern Cape, Free State, Gauteng, KwaZulu-Natal, Limpopo, Mpumalanga, Northern Cape, North West, and Western Cape.

Batloung Fruits currently sells apples, bananas, oranges, mangoes, and grapes.

Fruit	**Source Provinces**
Apples	Western Cape, Free State, Mpumalanga

Bananas	Limpopo, KwaZulu-Natal, Mpumalanga
Oranges	Limpopo, Mpumalanga, Eastern Cape
Mangoes	Limpopo, Mpumalanga, KwaZulu-Nata
Grapes	Western Cape, Northern Cape, Limpopo

Fruit Box Sizes

A major requirement of the business is that fruits are sold to street vendors according to different box sizes. Batloung Fruits uses three box sizes, as shown below.

Small	A small quantity of fruit intended for street vendors with limited capital or lower expected daily demand.
Medium	A medium quantity suitable for street vendors with moderate customer demand.
Large	A larger quantity intended for street vendors who purchase higher volumes.

The actual quantity and selling price may differ according to the type of fruit and box size.

The system must therefore distinguish between 'Fruit + Box Size'.

Apples	Small
Apples	Medium
Apples	Large

These should be treated as different saleable product configurations.

Street Vendors

The system should maintain information about each vendor, including Vendor ID, Vendor name, Business name, Telephone number, Location or area in Maseru, Registration date, and Status.

Business Problem

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Batloung Fruits currently faces the problems below

- Difficulty tracking fruit stock
- Difficulty identifying which supplier supplied particular fruits
- Difficulty monitoring purchases from South African provinces
- Difficulty recording street vendor orders
- Difficulty distinguishing small, medium and large boxes
- Difficulty calculating the total value of orders
- Difficulty determining which fruits are selling most frequently
- Difficulty monitoring outstanding orders
- Difficulty generating sales reports
- Risk of inaccurate or duplicated manual records

The proposed system must address these problems using a database-driven full-stack web application.

Assignment Objective

In groups, design and develop a Batloung Fruits Distribution Management System using React.js for the frontend, Node.js for the server-side runtime, Express.js for the REST API, and MySQL for data storage. The application must demonstrate how a modern web application can connect a React frontend to an Express/Node.js backend and a relational MySQL database.

Required System Features

Frontend	Dashboard	• Total fruits • Total suppliers • Total registered vendors • Available stock • Total orders • Pending orders

		• Completed orders • Sales value Use appropriate cards, tables, charts or visualizations
	Fruit Management	The user should be able to add a fruit, view fruits, edit fruit information, delete or deactivate a fruit, search for fruits, and filter fruits. • The system must include, apples, bananas, oranges, mangoes, and grapes.
	Box Size Management	The application must support small, medium, and large box sizes.
	Supplier Management	Create a supplier management interface allowing authorized users to add suppliers, view suppliers, edit suppliers, delete suppliers, search suppliers, and filter suppliers by province.
	Stock Management	The system must allow Batloung Fruits to record incoming stock and automatically update available stock where appropriate
	Street Vendor Management	Create a vendor management module allowing registration of vendors, view vendors, editing vendor information, and searching vendors.

	Order Management	A street vendor should be able to place an order for fruits. An order should include street vendor name, order date, fruit name, box size quantity of boxes, unit price, subtotal, total order amount, and order status
	Order Status	• Pending • Confirmed • Processing • Ready for Collection/Delivery • Completed • Cancelled
	Search and Filtering	Search and filter by fruits, street vendors, suppliers, orders, provinces, box sizes, and order statuses
Backend	Fruits	GET /api/fruits GET /api/fruits/:id POST /api/fruits PUT /api/fruits/:id DELETE /api/fruits/:id
	Suppliers	GET /api/suppliers GET /api/suppliers/:id POST /api/suppliers PUT /api/suppliers/:id DELETE /api/suppliers/:id
	Street Vendors	GET /api/vendors GET /api/vendors/:id

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		POST /api/vendors PUT /api/vendors/:id DELETE /api/vendors/:id
	Stock	GET /api/stock POST /api/stock PUT /api/stock/:id
	Orders	GET /api/orders GET /api/orders/:id POST /api/orders PUT /api/orders/:id DELETE /api/orders/:id
Database	Tables	Suppliers, fruits, box_sizes, products, street_vendors, purchases, orders, order_items
	Relationships	Supplier → Stock/Purchases Fruit → Products Box Size → Products Street Vendor → Orders Orders → Order Items Products → Order Items • Implement ERD
	Business Rules	• Only the five specified fruits may initially be sold • Every product must have a box size. • The available box sizes are Small, Medium and Large

		• A product must have a selling price • An order must belong to a registered street vendor • An order must contain at least one order item • The quantity ordered must be greater than zero • A street vendor cannot order more boxes than are available in stock • When an order is successfully completed, the relevant stock should be reduced • Cancelled orders should not reduce available stock • Supplier records must identify the South African province from which fruit was sourced • The system should prevent invalid or missing required information • Prices and totals must be calculated consistently • The system should maintain appropriate relationships between database records
	Reports	• Available Stock Report • Sales by Fruit

		• Sales by Box Size • Sales by Street Vendor • Purchases by South African Province • Pending Orders • Completed Orders • Total Sales by Date
Security; RBAC	Administrator	• Manage suppliers • Manage fruits • Manage box sizes • Manage stock • Manage street vendors • Manage orders • View reports
	Sales	View products Register street vendors Create orders View orders Update order status according to assigned permissions

Marking Rubric

Criteria		Mark
Frontend	React application structure and componentisation	5
	Navigation and routing	3
	Dashboard	4

	Fruit and box-size management	4
	Supplier and vendor management	4
	Order management interface	5
	Form validation, search and filtering	3
	Responsive UI and usability	2
Backend	Node.js/Express project structure	3
	RESTful API design	6
	CRUD operations	5
	Server-side validation	3
	Error handling and HTTP status codes	3
	Authentication/authorisation	3
	API security and code quality	2
Database	Database design	4
	ERD and relationships	4
	Tables, primary keys and foreign keys	4
	SQL queries and CRUD operations	3
	Data integrity and constraints	3
Full-Stack Integration	React–Express integration	3
	Express–MySQL integration	3
	End-to-end functionality	2
	Stock and/or order business rules	2
	Technical documentation	3

Documentation and Presentation	System diagrams	2
	GitHub	1
	Demonstration	2
	Individual contribution and ability to explain the system	2
TOTAL		**93**